

M - closed, oriented 3-manifold

Problem 3.63 Are the Chern-Simons invariants $CS(M)$ of M always rational?

The Invariant: Comes from gauge theory.

Let A be a connection on $M \times SU(2)$,
i.e.

$$A \in \Omega^1(M, \mathfrak{su}(2)) \cong \Gamma(\text{Hom}(TM, M \times \mathfrak{su}(2)))$$

$$\text{So } A = \sum_{i=1}^3 \alpha^i \sigma_i, \quad \sigma_1, \sigma_2, \sigma_3 = \text{Pauli matrices} \\ \Omega^1(M) \quad \text{spanning } \mathfrak{su}(2)$$

The Curvature of A is $F_A := dA + A \wedge A$.

A is flat if $F_A = 0$.

Defⁿ $CS(A) = \int_M \text{Tr} \left(dA \wedge A + \frac{2}{3} A \wedge A \wedge A \right)$

Why?? Shing-Shen Chen & Jim Simons '74

found them in attempt to find a combinatorial formula for characteristic classes of 4-manifolds.

They got stuck w/ boundary terms:

Fix $A \in \text{Conn}(M)$ ↙ Corrections on $M \times \text{SU}(2)$

Suppose $M = \partial W^4$, let $\begin{matrix} E \\ \downarrow \\ W \end{matrix}$ an $\text{SU}(2)$ -bundle
& $\tilde{A} \in \text{Conn}(E)$ an extension of A .

Then

$$\int_{M=\partial V} \text{Tr}(dA \wedge A + \frac{2}{3} A \wedge A \wedge A) = \int_M \text{Tr}(F_A \wedge F_A)$$
↙ Stokes' theorem

$$\overset{\text{Chern-Weil theory}}{\longrightarrow} = C_2(E)$$

More generally, this shows that

$CS(A_1) - CS(A_2) \in \mathbb{Z}$, for A_1, A_2 differing by a cob.

Bundle Automorphisms

$$\text{Aut}_{\text{Bun}}(M \times \text{SU}(2)) \hookrightarrow \text{Conn}(M)$$

$$\xrightarrow{\text{S11}} \text{Hom}(M, \text{SU}(2))$$

"Gauge group"

$$\& \text{ for } g \in \text{Hom}(M, \text{SU}(2)),$$

$$CS(g \cdot A) = CS(A) + \deg(g),$$

Fact

$$\left\{ \begin{array}{l} \text{Flat connections} \\ \text{on } M \times \text{SU}(2) \end{array} \right\} / \text{Gauge group} \xleftrightarrow{1:1} \left\{ \begin{array}{l} \text{homomorphisms} \\ \text{Tr}_1(M) \rightarrow \text{SU}(2) \end{array} \right\} / \text{Cons.}$$

!!
 $\mathcal{X}(M)$ "SU(2)-Character Variety"

Thus, there's a well-defined map:

$$CS: \mathcal{X}(M) \rightarrow \mathbb{R}/\mathbb{Z}$$

Defⁿ

$$CS(M) := \text{Im}(CS: \mathcal{X}(M) \rightarrow \mathbb{R}/\mathbb{Z})$$

Invariant shows up everywhere:

1) Instanton Floer homology

"Morse homology of Chern-Simons functional" - A 3+1 TQFT

A bunch of homology theories, but original is Floer's OG theory

IH : $\{\mathbb{Z}HS^3s\} \rightarrow \mathbb{Z}/8$ - graded vector spaces

Chain complex generators are flat connections
Admits a filtration built from CS invariants.

2) (Witten '89) The Jones polynomial
can be defined using the Chern-Simons
form. ↳ Won a Fields medal!

Also is categorified by (a version of) instanton
homology

3) Hyperbolic geometry - Complex volume is $CS(m) + i \text{vol}(m)$
↗ quantum invariants,
↖ volume conjecture

4) Number theory ??? (arithmetic Chern-Simons)

Rationality - what is known (Notably, all rational or unknown)

• (Furuta '87, Fintuskel - Stern '88) -

CS for Seifert fibered $\mathbb{Z}HS^3s$

(to show $\exists$ family of lin. indep. $\mathbb{Z}HS^3s$ in

the homology cobordism group - this implicitly relies on the filtered structure of instanton homology)

- (Kirk-Klassen '90) Gave a formula for CS in terms of area between curves on "the pillow case" - $\chi(S \times S')$ for 3-mflds w/ torus boundary

Computed CS for

- i) Lens spaces
- ii) Seifert Fibered $\mathbb{Z}HS^3_S$ (again)
- iii) Small surgeries on the figure-eight knot^(*)
- iv) "Binary dihedral" reps of mapping tori of the torus

- (Auckly '91, '92) Used KK formula to compute CS for

- i) All Seifert fibered manifolds
(including non-orientable base)
- ii) All graph manifolds
- iii) Binary dihedral reps for all 3-mflds which fiber over S^1

iv) Special reps of all 3-mflds

↑
"Component-defined"

Open (Problem 3.104) Are all reps special?

Essentially left w/ hyperbolic case.

- (Hilden et al. '99) Hyperbolic 3-mflds obtained as virtually regular coverings of S^3 w/ a geometric orbifold structure compatible w/ an amphichiral, hyperbolic knot have rational CS invariants. (whew...)

(*) Via numerical approximation — not all CS values are known to be rational.

Potential Counter-Examples

- $\frac{1}{2}$ Surgery on figure-eight
- $\frac{1}{2}$ Surgery on 5_2
- Surgery on other small hyperbolic 2-bridge knots

Approximated to 50+ digits, doesn't look rational...

Problem 3.104 (part 2) Does every element of $CS(M)$ come from one on a connected sum of lens spaces?

Implications of (Ir)rationality

Has implications in

- Instanton homologies
- (Homology) cobordism obstruction
- Quantum topology
(Reshetikhin-Turaev invariants / volume conjecture)
- Algebraic K -theory
- Regulators & special functions of L -functions
- The Bloch group
- etc...